Quantification: 701_z26c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	45
interior lacuna count	33
frame-edge lacuna count	12

Interior lacunae only: mean, median and sample SD (n = 33)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	2059.24	1422	2588.69	px^2
major_axis_length_px	lacuna shape	74.114	63.8935	37.7848	px
minor_axis_length_px	lacuna shape	36.4083	30.2759	21.4604	px
aspect_ratio	lacuna shape	2.1894	2.0618	0.7626	unitless
eccentricity	lacuna shape	0.8239	0.8745	0.1557	unitless
solidity	lacuna shape	0.7404	0.7421	0.0904	unitless
roots_count	headline	7.6364	7	4.0606	count
ring_length_r30_px	headline	286.061	267	146.981	px
ring_length_r60_px	reported	553.424	549	340.473	px
owned_length_px	ownership-dependent	723.295	468.262	670.729	px
edge_count	ownership-dependent	49.7273	34	45.5064	count
mean_edge_length_px	ownership-dependent	14.7909	14.5337	2.9287	px
perimeter_px	normalised	232.694	167.823	179.157	px
ring_area_r30_px2	normalised	5879.42	4822	3321.16	px^2
ring_area_r60_px2	normalised	11122.2	9318	6552.79	px^2
in_frame_fraction_r30	normalised	0.9984	1	0.0095	unitless
in_frame_fraction_r60	normalised	0.9944	1	0.0257	unitless
ring_density_r30	normalised	0.0511716	0.0471634	0.013647	px^-1
ring_density_r60	normalised	0.0512591	0.047405	0.0142459	px^-1
roots_per_100px_perimeter	normalised	4.0413	3.7089	1.7948	per 100 px
ring_attached_length_r30_px	network v2	246.788	213	134.557	px
ring_attached_length_r60_px	network v2	413.212	333	286.517	px
ring_length_w_r30_px	network v2	322.084	310.551	166.079	px
ring_length_w_r60_px	network v2	622.63	612.115	384.435	px
sholl_crossings_r10	network v2	7.3333	6	4.0671	count
sholl_crossings_r20	network v2	7.3636	7	4.2265	count
sholl_crossings_r30	network v2	7.1515	7	4.8934	count
loop_count	loops, report only	2.7879	1	5.2782	count
loop_edge_share	loops, report only	0.1257	0.0732	0.1546	share
ring_width_mean_r30_px	width	6.2423	6.4017	1.38	px

Per field
----

measure	value	units
analysed_area_px2	895541	px^2
total_skeleton_length_px	43791	px
canalicular_length_density_per_px	0.0488989	px^-1
junction_count	1271	count
junction_density_per_px2	0.00141925	px^-2
lacuna_count	45	count

lacunae_per_px2	5.025e-05	px^-2
skeleton_component_count	688	count
mean_component_length_px	63.6497	px
median_component_length_px	14	px
field_length_density_w_per_px	0.05499	px^-1
field_density_without_flagged_per_px	0.0497959	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	286	px
unattached_component_count	16	count
canalicular_length_density_attached_per_px	0.0488989	px^-1
field_length_density_with_unattached_per_px	0.0492183	px^-1
loop_count	162	count
loop_edge_share	0.2093	share
edge_count	3215	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.5351	px
width_median_px	6.3246	px
width_p10_px	2.8284	px
width_p90_px	10.198	px
width_pixels_used	29501	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	6622	126.492	79.7855	1.5854	0.776	0.7588	0.7952	43.4012	272.285	True
2	24115	285.31	118.544	2.4068	0.9096	0.7591	1.2945	92.9897	132.363	True
3	8121	199.992	79.1327	2.5273	0.9184	0.6393	0.943	189.47	54.1918	True
4	6792	139.701	72.9543	1.9149	0.8528	0.7645	1.2651	31.6926	963.876	True
5	2287	125.157	32.9459	3.7989	0.9647	0.6065	0.5498	75.4101	463.222	False
6	3999	122.318	44.0398	2.7774	0.9329	0.877	0.1379	179.418	1003.78	True
7	1531	85.4004	40.5987	2.1035	0.8798	0.5916	1.0809	183.008	698.054	False
8	2493	118.127	38.0041	3.1083	0.9468	0.6054	-1.3995	230.773	666.071	False
9	5110	133.797	67.4988	1.9822	0.8634	0.6757	1.2061	267.96	890.703	False
10	1782	62.6142	47.9495	1.3058	0.6431	0.682	1.1297	243.333	206.608	False
11	459	45.5099	13.5856	3.3499	0.9544	0.8438	0.9355	286.754	149.582	False
12	1422	68.8124	28.5123	2.4134	0.9101	0.8335	1.1259	314.225	198.443	False
13	1652	67.0082	31.8663	2.1028	0.8797	0.8959	0.9049	359.245	256.938	False
14	2608	67.6663	54.9533	1.2313	0.5835	0.8644	0.6706	338.284	24.3976	True
15	30788	334.972	147.951	2.2641	0.8972	0.683	0.7858	446.787	150.129	True
16	12693	158.381	111.247	1.4237	0.7118	0.8419	-1.4075	620.143	187.231	False
17	967	67.0717	21.9089	3.0614	0.9451	0.7662	-1.3191	319.159	644.63	False
18	8889	153.266	79.6369	1.9246	0.8544	0.9019	1.3702	455.699	960.443	True
19	821	63.8935	20.0569	3.1856	0.9495	0.7609	0.1039	594.211	741.577	False
20	4845	153.224	58.0129	2.6412	0.9256	0.6393	-0.5	676.391	794.742	False
21	2272	79.8087	42.6154	1.8728	0.8455	0.7881	-1.0855	686.797	269.451	False
22	640	42.0108	20.3754	2.0618	0.8745	0.8625	1.0676	716.059	885.861	False
23	656	33.1626	30.2759	1.0953	0.4081	0.801	1.2644	753.977	863.463	False
24	883	72.1056	21.4511	3.3614	0.9547	0.6761	-1.2791	751.882	124.115	False

25	1013	51.2657	31.9563	1.6042	0.7819	0.7251	-0.0492	792.697	1003.49	True
26	1435	61.7195	33.8344	1.8242	0.8363	0.8112	-1.4959	787.26	645.571	False
27	4376	111.203	63.8724	1.741	0.8186	0.6554	-1.0966	845.546	482.61	False
28	633	36.1911	26.3602	1.3729	0.6852	0.7421	-1.1058	875.441	532.795	False
29	474	32.816	22.1225	1.4834	0.7386	0.7732	0.5755	897.675	908.236	False
30	1500	55.4417	51.2017	1.0828	0.3835	0.6268	-1.2338	890.901	936.497	False
31	405	33.2285	16.2908	2.0397	0.8716	0.8544	-1.2783	904.956	141.983	False
32	718	51.4892	21.7937	2.3626	0.906	0.7349	-1.184	917.613	557.102	False
33	563	40.0099	23.9496	1.6706	0.8011	0.7136	0.7843	936.179	478.274	False
34	864	54.6671	23.8765	2.2896	0.8996	0.7279	-1.1377	941.147	439.961	False
35	3941	116.706	45.969	2.5388	0.9192	0.89	-1.4794	1004.56	375.487	True
36	2384	124.611	32.5415	3.8293	0.9653	0.6932	-1.5289	1010.89	775.227	True
37								3.9238	660.128	True
38								215.421	812.396	False
39								313.729	249.554	False
40								622.916	889.099	False
41								636.105	618.971	False
42								664.2	750.446	False
43								691.745	622.696	False
44								885.169	566.925	False
45								937.292	398.788	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	15	344	618	349.421	24	14.5592	5.9536
2	26	791	905	814.482	66	12.3406	4.6531
3	10	327	384	217.995	17	12.8232	5.9014
4	14	492	732	432.676	39	11.0943	5.4987
5	19	555	1192	748.027	57	13.1233	6.735
6	11	382	548	859.11	51	16.8453	6.1919
7	11	360	565	361.149	28	12.8982	5.9174
8	12	464	820	1124.78	72	15.6219	5.0033
9	15	515	1038	1178.56	84	14.0305	6.5919
10	9	291	334	261.995	16	16.3747	5.9108
11	5	117	120	179.924	10	17.9924	6.0585
12	6	113	118	64.2843	6	10.714	3.8149
13	4	166	336	96.1543	5	19.2309	4.3661
14	1	27	84	7.8284	1	7.8284	9.1513
15	12	520	983	330.563	24	13.7735	4.4296
16	12	382	711	794.014	54	14.704	5.4258
17	9	294	549	412.191	32	12.881	6.4769
18	7	469	1252	1648.74	109	15.126	5.4301
19	5	267	564	846.039	58	14.5869	7.2624
20	14	545	863	680.299	47	14.4744	4.4587
21	8	333	825	1203.98	77	15.6361	6.7465
22	5	173	311	146.569	17	8.6217	5.4383
23	7	190	292	277.723	22	12.6238	5.4386
24	8	486	1127	2112.94	155	13.6319	7.8175
25	6	190	339	235.48	12	19.6234	5.8946
26	8	487	1040	1758.58	121	14.5337	7.4873
27	16	566	1275	2071.21	150	13.808	5.9685
28	6	130	132	209.468	14	14.962	3.118
29	4	154	421	682.215	41	16.6394	6.4213
30	8	345	658	363.22	23	15.7922	6.3822
31	5	277	924	2659.97	173	15.3756	6.8562
32	8	358	698	1543.69	97	15.9143	8.3854
33	7	253	379	410.12	30	13.6707	8.9378

34	6	206	210	166.811	14	11.9151	
35	15	355	629	1089.44	70	15.5634	4.193
36	9	276	681	96.3553	7	13.765	
37	2	88	309	861.365	54	15.9512	4.9484
38	2	121	245	378.907	25	15.1563	7.4481
39	2	98	196	26.5563	1	26.5563	4.2418
40	5	291	665	1268.51	78	16.263	7.5127
41	6	242	571	468.262	34	13.7724	6.3442
42	3	102	122	150.125	11	13.6477	4.6904
43	7	190	283	247.409	15	16.4939	7.7015
44	3	112	251	480.22	39	12.3133	6.8794
45	7	257	428	494.848	35	14.1385	7.9174

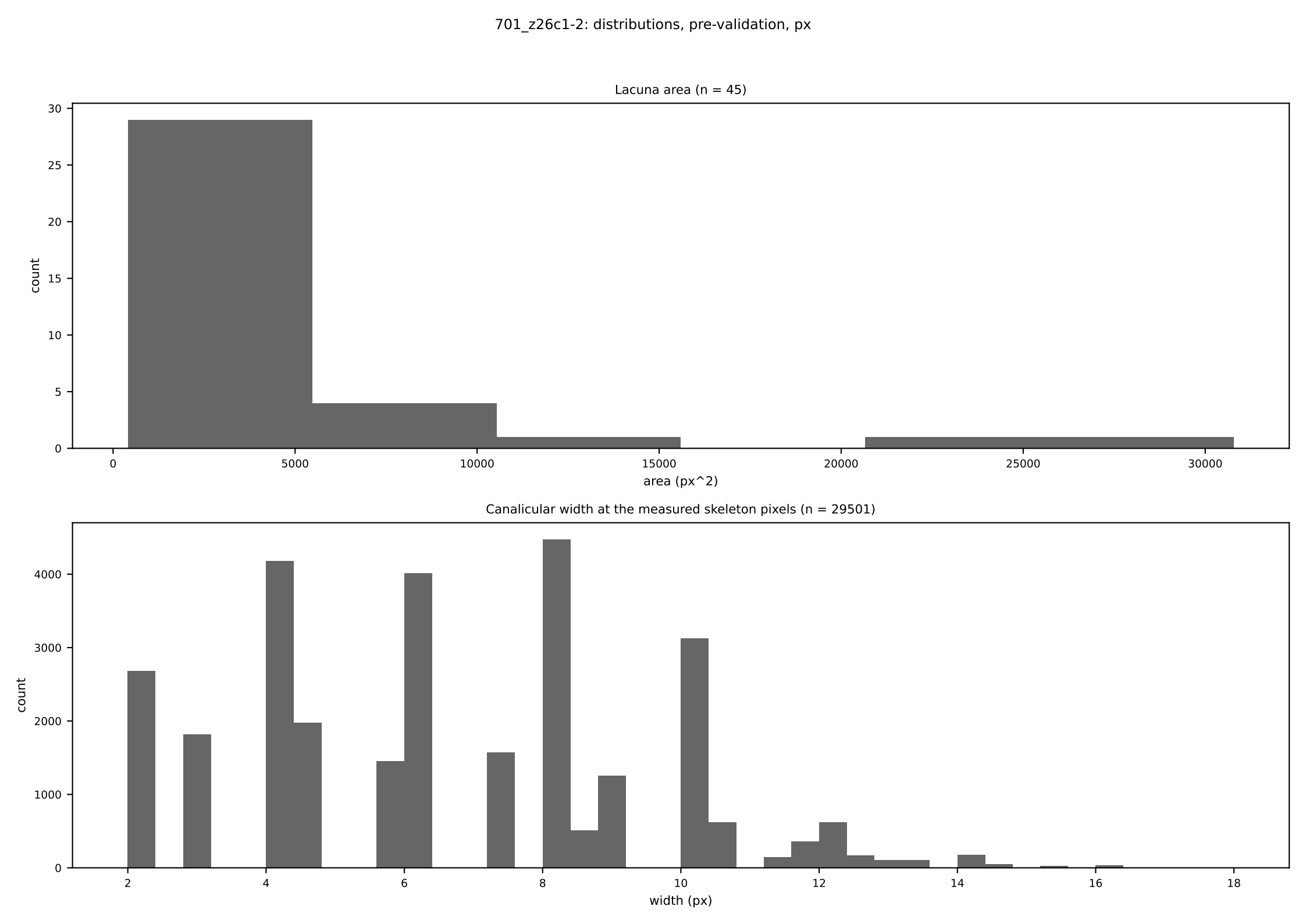
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	751.134	8143	12926	0.7412	0.6831	0.0422449	0.0478106	1.997
2	1296.59	17823	20380	0.8324	0.6918	0.0443809	0.0444063	2.0053
3	652.802	9448	10851	0.7803	0.7374	0.0346105	0.0353884	1.5319
4	561.257	7623	12937	0.5264	0.4563	0.0645415	0.0565819	2.4944
5	459.683	11984	25145	0.9457	0.8575	0.0463117	0.047405	4.1333
6	376.019	7018	10546	0.6142	0.5806	0.0544315	0.0519628	2.9254
7	322.475	7086	13868	1	1	0.0508044	0.0407413	3.4111
8	439.713	9174	15810	1	1	0.0505777	0.0518659	2.7291
9	541.524	13241	24654	1	1	0.0388943	0.0421027	2.77
10	345.031	6611	9095	1	1	0.0440176	0.0367235	2.6085
11	115.539	2499	2647	1	1	0.0468187	0.0453343	4.3275
12	202.35	3335	3424	1	1	0.0338831	0.0344626	2.9652
13	196.522	5138	8606	1	1	0.0323083	0.0390425	2.0354
14	273.48	2279	3229	0.6705	0.6272	0.0118473	0.0260142	0.3657
15	1396.63	22077	31992	0.9271	0.8877	0.0235539	0.0307264	0.8592
16	659.572	11568	19301	1	1	0.0330221	0.0368375	1.8194
17	188.244	7262	16365	1	1	0.0404847	0.0335472	4.781
18	546.701	11662	25518	0.7611	0.7175	0.0402161	0.0490634	1.2804
19	163.723	6093	12577	1	1	0.0438208	0.0448438	3.0539
20	660.199	12837	18982	1	1	0.0424554	0.0454641	2.1206
21	273.019	7642	15822	1	1	0.043575	0.0521426	2.9302
22	128.202	4822	8070	1	1	0.0358772	0.0385378	3.9001
23	133.338	4666	9184	1	1	0.0407201	0.0317944	5.2498
24	277.948	7921	17655	1	1	0.0613559	0.0638346	2.8782
25	208.386	5088	9797	0.7064	0.6319	0.0373428	0.0346024	2.8793
26	215.7	7866	16779	1	1	0.061912	0.0619822	3.7089
27	605.222	10588	19289	1	1	0.0534567	0.0660999	2.6437
28	146.202	2096	2101	1	1	0.0620229	0.0628272	4.1039
29	133.717	2752	7041	1	1	0.0559593	0.0597926	2.9914
30	307.298	7315	14828	1	0.9574	0.0471634	0.0443755	2.6033
31	96.7107	5294	16215	1	1	0.0523234	0.0569843	5.1701
32	167.823	4472	9318	1	1	0.0800537	0.0749088	4.7669
33	143.616	3510	5238	1	1	0.0720798	0.0723559	4.8741
34	186.279	2557	2589	1	1	0.0805632	0.0811124	3.221
35	332.007	5720	10294	0.5904	0.5692	0.0620629	0.0611036	4.518
36	413.233	6749	15845	0.5979	0.5709	0.0408949	0.0429789	2.178
37	95.9411	2922	8668	0.5282	0.5187	0.0301164	0.0356484	2.0846
38	51.598	2879	5988	1	1	0.0420285	0.0409152	3.8761
39	57.9411	2244	3907	1	1	0.043672	0.0501664	3.4518
40	78.5685	4732	10434	1	1	0.0614962	0.0637339	6.3639
41	82.5685	4641	11473	1	1	0.0521439	0.049769	7.2667
42	70.4264	2115	3032	1	1	0.0482269	0.0402375	4.2598

43	68.2193	4210	8536	1	1	0.0451306	0.0331537	10.261	
44	70.669	1878	3542	1	1	0.0596379	0.0708639	4.2451	
45	89.2548	2993	5516	1	1	0.085867	0.0775925	7.8427	

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	304	368	379.563	684.215	8	5	8
2	697	772	886.838	1019.55	18	25	5
3	240	240	368.735	432.777	11	9	1
4	434	457	561.76	830.382	14	9	12
5	494	692	631.843	1347.07	20	10	12
6	316	488	430.09	617.357	10	15	7
7	283	315	409.534	633.973	9	8	9
8	407	569	519.032	917.21	13	12	8
9	481	745	589.914	1171.12	15	17	15
10	192	217	318.38	365.007	7	7	2
11	102	102	132.397	135.225	2	2	0
12	77	77	121.669	128.326	4	1	2
13	101	125	188.066	379.262	5	7	4
14	8	8	30.799	97.9117	1	0	3
15	287	295	583.085	1099.74	10	14	13
16	315	453	429.505	793.997	10	8	9
17	218	321	326.936	612.115	6	7	7
18	379	801	519.475	1402.79	14	16	13
19	213	447	310.735	644.156	6	6	10
20	413	619	605.458	971.092	12	14	9
21	320	575	374.492	920.997	8	9	11
22	125	125	189.125	341.108	4	7	8
23	190	240	211.095	317.622	5	7	2
24	461	1057	552.404	1284.22	9	14	14
25	146	192	210.167	381.333	5	3	6
26	448	926	543.576	1164.47	11	15	19
27	512	1110	631.831	1417.18	14	13	14
28	120	120	145.811	148.811	6	2	0
29	154	333	176.539	479.818	3	5	4
30	262	368	394.676	736.855	9	9	13
31	244	802	310.551	1049.19	7	8	10
32	353	541	398.664	795.441	9	10	6
33	208	287	291.794	434.191	6	8	4
34	206	210	233.439	237.267	6	2	2
35	317	564	387.593	701.872	9	8	8
36	107	107	303.007	755.269	5	5	13
37	61	107	98.8406	353.12	2	2	5
38	121	237	132.083	274.309	2	4	3
39	47	47	108.841	219.853	2	3	1
40	268	653	342.806	781.11	5	7	12
41	191	373	270.894	639.044	5	9	9
42	94	94	116.154	138.468	5	1	1
43	178	205	214.095	315.279	5	6	5
44	101	242	124.598	273.539	4	2	5
45	245	409	281.853	479.463	8	3	6


701_z26c1-2: distributions, pre-validation, px
Lacuna area (n = 45)
count
area (px^2)
0
5
10
15
20
25
30
0
5000
10000
15000
20000
25000
30000
Canalicular width at the measured skeleton pixels (n = 29501)
count
width (px)
0
1000
2000
3000
4000
2
4
6
8
10
12
14
16
18